Student Name: Michal Pokojny
Student ID : x23355999
Github Repository: https://github.com/MichalPokojny/smart-docklands-NCI-Project-x2335599
Date: 13/03/26

National College of Ireland

Project Submission Sheet

Student Name:	Michal Pokojny		
Student ID:	X23355999		
Programme:	Computing	**Year:**	2025/26
Module:	Data Structures & Algorithms		
Lecturer:	Arjun Vinayak Chikkankod		
Submission Due Date:	18/03/2026		
Project Title:	Data Structures & Algorithms CA Project		
Word Count:	650		

Academic Integrity Certificate Verification Code: APzTxezPXB

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.
ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard or appropriate referencing style supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students

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may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Note: Academic Integrity Course Requirement – Starting **Monday, 19th January 2026**, **all students are required to successfully complete the Epigeum Academic Integrity Course**, which is available on **Moodle**. This is a **self-paced online course** consisting of **4 modules** and will take approximately **2 hours** to complete. Once you have successfully completed the course, you will receive a **Certificate & Verification Code**. This code **must be included on the Project Cover Sheet** when submitting your work. For full details and guidance, please visit: 📎 ttps://libguides.ncirl.ie/academicintegrity/epigeumforstudents

Signature: Michal Pokojny

Date: 26/02/2026

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
6. **Please check that you read AI and Academic Integrity Acknowledgement Supplments in this document**

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Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

Data Structures & Algorithms

Online CA Project

Your Name/Student Number	Course	Date
Michal Pokojny/x23355999	Computing – Data Structures & Algorithms	13/03/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click here.

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
ChatGPT (OpenAI)	**Brainstorming application theme and ideas**	https://chatgpt.com/

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

ChatGPT

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Can you help me brainstorm ideas for my simple Java app project, that fits for the theme on this website https://smartdocklands.ie/smart-docklands-call-for-pilots-2025-deadline-extended/ I am thinking of creating app that fits the theme : Urban Greening / Biodiversity	**The programme organised by Smart Docklands encourages innovative technology ideas that improve urban life in Dublin. One of the main themes in the Smart Docklands Call for Pilots 2025 is Urban Greening / Biodiversity, which focuses on solutions that support green infrastructure, environmental monitoring, and community engagement with nature in cities.** **For a simple Java Swing project, the best ideas are those that:** • **use simple classes and objects** • **store data in lists or files** • **have a GUI with forms, buttons, and tables** • **perform simple calculations or tracking.** **Below are several ideas that fit the theme and would work well as a basic Java application.**
	1. Urban Tree Tracker 🌳 **Concept** **An application that tracks trees in a city area and their health.** **Features** **Users can:** • **Add a tree (species, location, date planted)** • **Update tree condition (healthy, needs maintenance, damaged)** • **View a list of all trees** • **Filter by species or health status** **Example classes** • **Tree** • **TreeManager** • **Location** **GUI** • **Add Tree form**

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- **Tree List table**
- **Tree Details panel**

Why it fits the theme

Trees are essential for urban greening, cooling cities, and supporting biodiversity.

---

2. Biodiversity Sightings Logger 🦜

Concept

A simple app where people record wildlife or plant sightings in urban areas.

Users record

- **species name**
- **category (bird, insect, plant, mammal)**
- **location**
- **date observed**

Features

- **Add new sightings**
- **View all sightings**
- **Show statistics (e.g., most common species)**

Example classes

- **Sighting**
- **Species**
- **ObservationManager**

GUI

- **Add Observation form**
- **Observation List**
- **Statistics panel**

Why it fits

Encourages citizen participation in biodiversity monitoring.

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	3. Smart Waste Bin Monitoring App ♻️ **Concept** **A simple system for managing public waste bins in urban green spaces.** **Features** • **Store bin locations** • **Track bin fill level** • **Mark bins that need collection** • **View bins that are nearly full** **Example classes** • **SmartBin** • **BinManager** • **CollectionSchedule** **GUI** • **Add Bin** • **Bin Status List** • **Collection Priority view** **Why it fits** **Better waste management helps protect green spaces and wildlife from pollution.**
	✅ **Best options for a simple Java Swing project** 1. **Urban Tree Tracker** – easiest to implement 2. **Biodiversity Sightings Logger** – interesting and educational 3. **Smart Waste Bin Monitoring** – practical smart-city idea

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

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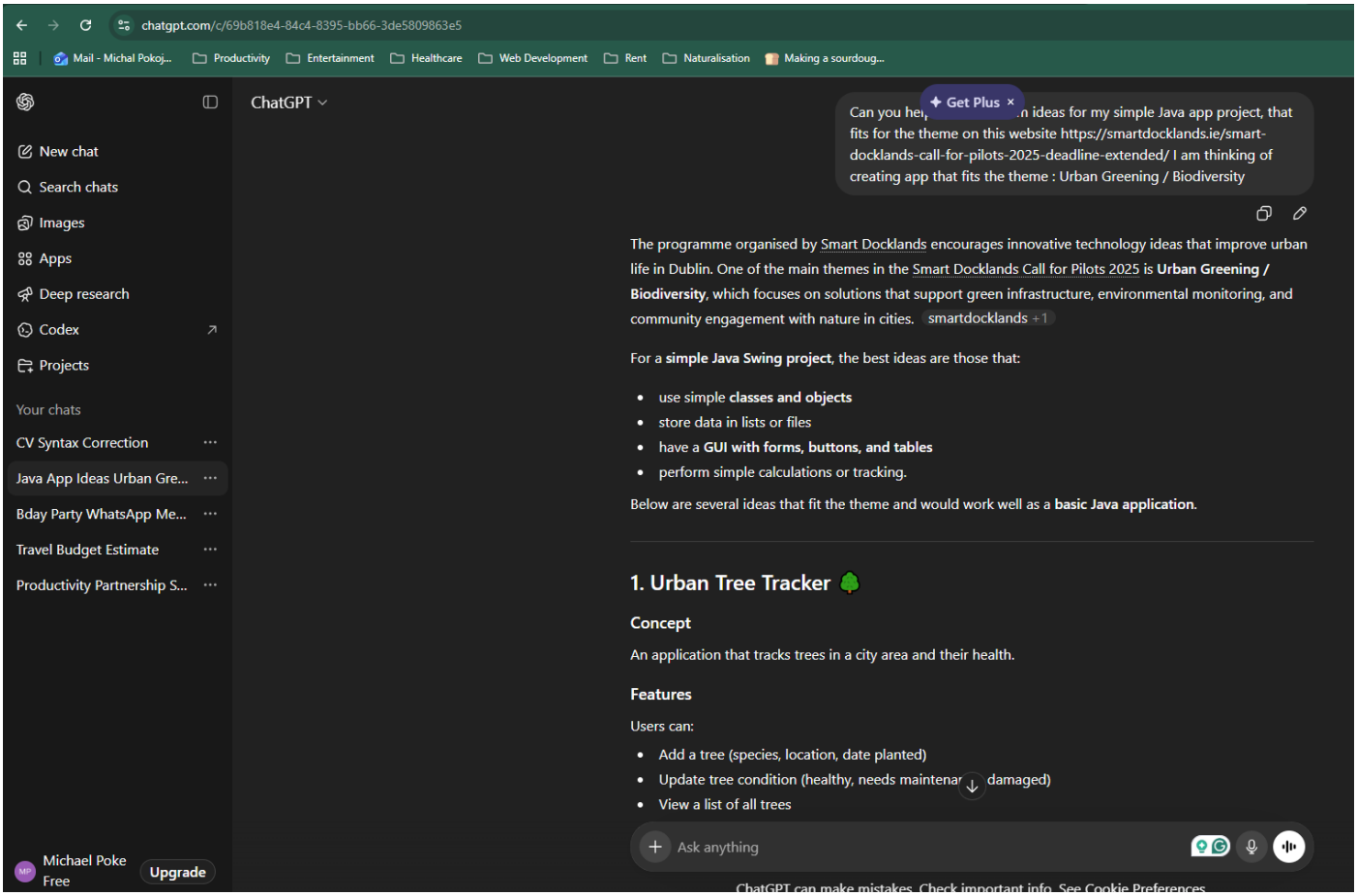
Additional Evidence:



Additional Evidence:

N/A

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Project Proposal

SmartWaste: Smart Waste / Bin Collection Management System

Theme: Urban Greening / Biodiversity - Smart Docklands Call for Pilots 2025

1. Selected Theme

This project is based on the *Smart Docklands Pilot Call Themes* focusing on smart waste management and environmental sustainability in the Dublin Docklands area. The Smart Docklands initiative represents a collaborative effort between Dublin City Council, local businesses, academic institutions, and technology partners to transform the Docklands into a living laboratory for smart city innovations.
This project specifically addresses the environmental sustainability, targeting waste management optimization as a key component of creating a cleaner, more efficient, and technologically advanced urban environment.

2. Application Description

The Smart Waste/Bin Collection System is a prototype desktop application developed in Java using the Swing GUI framework for user interaction that can be easily deployed into real-world use.

This application uses 3x Abstract Data Types (ADTs) taught in class to manage waste collection operations efficiently:
- Singly Linked List for managing optimized collection routes
- Stack for tracking maintenance history records
- Queue for handling incoming collection requests in a first-come, first-served manner.

The system provides a simple interface for registering smart bins across the Docklands area, monitoring their fill levels, processing citizen requests and maintaining detailed collection or service histories. Built using object-oriented programming principles such as inheritance, polymorphism, and interface implementation while solving waste management challenges through efficient data structure utilization.

3.Problem Solution

The Dublin Docklands area faces significant waste management challenges due to high population density, mixed commercial and residential usage, and increasing urbanization.

Current bin collection systems operate on fixed schedules regardless of actual bin capacity, leading to:
- Overflowing bins between collection cycles, creating health and environmental hazards

- Unnecessary collection trips for bins that are not full, wasting fuel and resources
- Lack of real-time monitoring and reporting capabilities
- Poor visibility into maintenance needs for damaged or malfunctioning bins

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These inefficiencies result in increased carbon emissions, higher operational costs for Dublin City Council, and degraded environmental quality in the Docklands area. The Smart Bin Collection System addresses these challenges by introducing intelligent monitoring, dynamic scheduling, and optimized collection schedules that adapt to actual bin usage patterns rather than relying on manually set collection schedules.

4.Alignment with Smart Docklands Theme

SmartWaste application supports the Smart Docklands goal of using technology to improve sustainability, city operations, and quality of life in the Docklands area.

It reduces fuel use and emissions by optimising collection routes and only scheduling trips when bins are close to full, supporting Dublin's climate targets.

By collecting and displaying bin and collection data, it enables more informed, data-driven decisions about waste infrastructure and resource allocation.

Using simulated sensor data and connected "smart" bins, the system acts as a practical example of smart city infrastructure in the Docklands.

Cleaner streets and more reliable bin collection improve the daily experience for residents, workers, and visitors, in line with the citizen-focused aims of Smart Docklands.

The modular, scalable design also makes it suitable as a base for future smart city services and partnerships between the council, industry, and the community.

5.Technical Overview

Technical components:

Inheritance and Polymorphism: An abstract class SmartWaste is inherited by 2x subclasses — GeneralWasteBin & RecyclingBin.
Each overrides the getCollectionPriority() method, demonstrating polymorphism.

Abstract Data Types : Three different ADTs are implemented, each backed by an interface defining its required behaviours:

- SinglyLinkedList - manages the bin collection route, with full CRUD operations
- Stack — stores maintenance history per bin (using LIFO – Last In First Out)
- Queue — manages collection requests in arrival order (FIFO – First In First Out), with full CRUD operations

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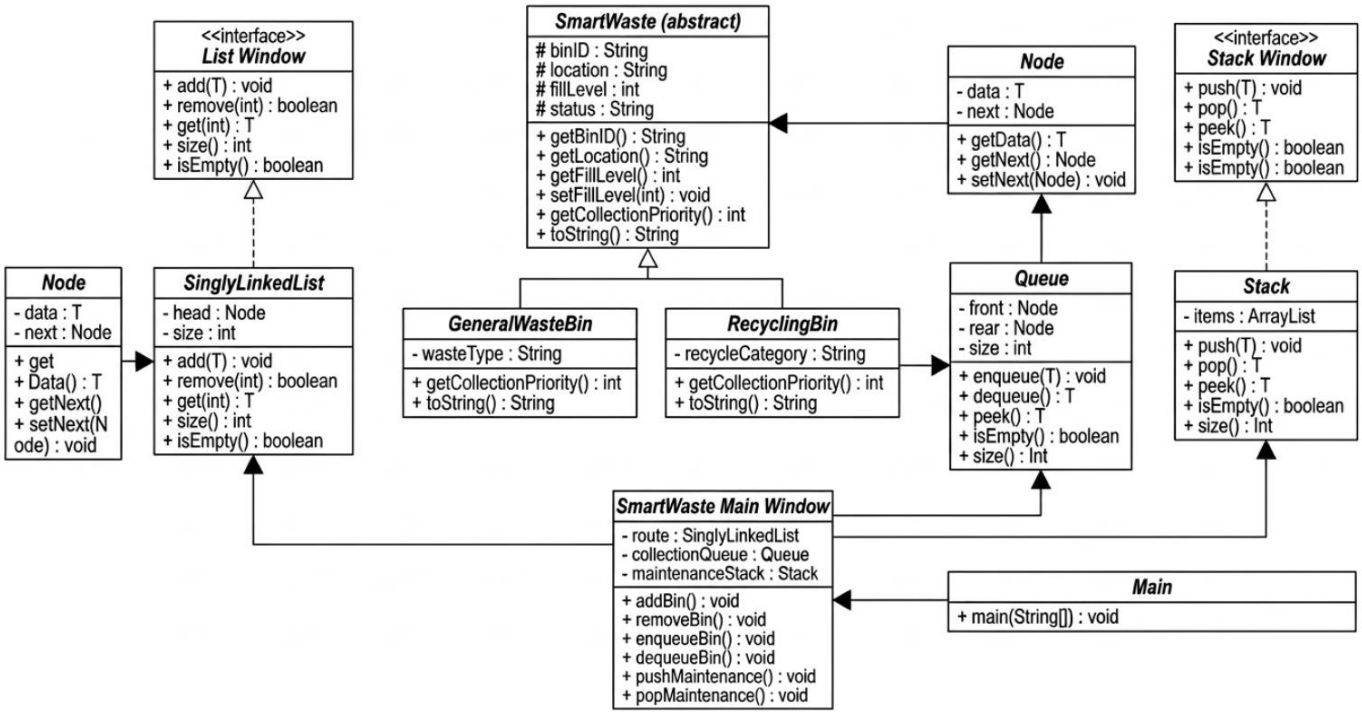
GUI: Built using Java Swing in NetBeans GUI Builder.

A JFrame with a JTabbedPane use JTextField, JComboBox, JButton, and JTable components with input validation before data is stored to any ADT.

Object Storage: All data entered through the GUI is stored as Java objects within the ADTs.

Class Diagram



Technologies used:

Java (JDK 25) — Programming language used to build the full application and all ADT implementations

Apache NetBeans 29 IDE — Development environment used to write, compile, run and design the Swing GUI forms

Java Swing GUI — Java toolkit providing all desktop interface components such as JFrame, JTable, and JButton

GitHub — Version control platform used to track and commit code throughout the project

draw.io — Used to design and create the UML class diagram for this proposal

ChatGPT (OpenAI) — Used for brainstorming the application ideas and theme only

Word count (Heading 1. - 4) 493 + 5. Heading 216